TANUSH THAKRAN

AI / ML Engineer

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SUMMARY

Machine learning engineer who ships models with evidence behind them: leakage-safe evaluation, controlled ablations, and metrics reported against honest baselines. Experience spans learning-to-rank on multi-million-row datasets, applied ML with scikit-learn, LLM/RAG systems, and full-stack web applications containerised with Docker, owning the full path from data extraction through deployment.

TECHNICAL SKILLS

Machine Learning: LightGBM, scikit-learn, Feature Engineering, Cross-Validation, Model Evaluation. **Familiar with:** PyTorch, TensorFlow

LLM & Retrieval: RAG, Embeddings, Vector Search (ChromaDB), Prompt Engineering, LLM Evaluation, Groq API, Gemini API, Whisper, Ollama

Data & Analytics: Python, SQL, PostgreSQL, Pandas, NumPy, PyArrow/Parquet, Data Cleaning, Statistics, Data Visualisation, Business Intelligence, Dashboards, Power BI, DAX, Power Query

Full-Stack & Infra: Full-Stack Web Application Development, REST APIs, FastAPI, Node.js, React.js, Next.js, Docker & Docker Compose, Celery, Redis, Streamlit, pytest, GitHub Actions, Git, Flutter

MACHINE LEARNING & DATA PROJECTS

Pokémon TCG Move Prediction — Learning-to-Rank

Python, LightGBM, PyArrow | *github.com/Tanush1206/pokemon-tcg-move-prediction*

- Framed move selection as learning-to-rank: an LGBMRanker with a LambdaRank objective trained on 3.35M rows (83 features) from 4,604 recorded battles, reaching 57.4% top-1 accuracy on 50,649 held-out decisions vs. a 21.7% random baseline (2.65x); the true move is in the top 3 88% of the time.
- Ran a 7-run ablation on a fixed test set with 70/15/15 episode-level splits: the largest gain (+0.054) came from reverse-engineering undocumented option indices to resolve which card each option refers to, while filtering for “stronger” episodes did not help (matched-volume control).
- Built a multiprocessing extraction pipeline: 4,604 game files streamed into one Parquet file in about 75 seconds on 31 workers.

Customer Churn Prediction — Deployed ML App

Python, scikit-learn, Streamlit | *github.com/Tanush1206/Customer-Churn-Prediction*

- Built an end-to-end pipeline on IBM Telco data (7,043 customers): leakage-safe ColumnTransformer; logistic regression, random forest and gradient boosting compared on held-out data; logistic regression kept for interpretability (0.003 ROC-AUC below gradient boosting) and tuned with 5-fold CV. Chose a 0.30 threshold from out-of-fold predictions, lifting churn recall from 55.9% to 75.1% at 51.8% precision (84.1% ROC-AUC).

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python, Whisper, Ollama | *github.com/Tanush1206/rag_based_ai*

- Built a fully offline RAG pipeline over Hindi-language course videos (ffmpeg, Whisper large-v2 transcription with translation to English, local bge-m3 embeddings, top-5 cosine retrieval, deepseek-r1 via Ollama); answers cite video, title and timestamp and say so when the answer is missing; zero external API calls; fixed a scale-only embedding crash with batched requests.

LLM Utility Lab

Python, Groq API, FastAPI, Docker | *github.com/Tanush1206/llm-utility-lab*

- Built a modular LLM service on the Groq API (FastAPI, Docker, deployed on Render) with token tracking, typed error handling and an evaluation runner that records model, temperature and prompt version and flags pass-to-fail regressions; 58 pytest tests in GitHub Actions CI.

Superstore Profitability Analysis — SQL & Power BI

PostgreSQL, SQL, Power BI, DAX, Power Query | *github.com/Tanush1206/superstore-powerbi-analysis*

- Analysed 9,994 line items ($2.3M revenue, 2014–2017) in PostgreSQL (window functions, cohorts, RFM) and Power BI: margin is 29.5% undiscounted, turns negative past a 25% discount and reaches −77% beyond 40%. Central region discounts 24.0% for a 7.9% margin vs. West’s 10.9% for 14.9%, a gap worth about $35K over four years; proposed a single-region test before rollout; reconciled the data load against control totals.

VideoCaptionMaker — Private, Offline AI Captioning App **(Released v1.2.2)**

Python, FastAPI, Celery, faster-whisper, ChromaDB, Next.js, Docker Compose | *github.com/Tanush1206/Video-Caption-Maker*

- Built and released a local-first captioning app: faster-whisper transcription and M2M100 translation run on the user’s own GPU or CPU in 9 languages including Hinglish, with a caption editor, semantic search and burned-in MP4 or SRT/VTT export.
- Shipped a one-command installer with SHA-256-verified downloads and hardware-aware model selection (large-v3 on GPU, turbo or small on CPU), plus crash recovery for interrupted jobs and a request guard against DNS rebinding; 490+ backend tests.
- Measured accuracy on Google’s FLEURS set (3–7% word error on European languages, 26% on Hindi): found a silence bug on quiet recordings (English 59% → 6% after the fix) and removed four languages scoring 63–122%.

EXPERIENCE

App Developer Intern — CYPARTA Dec 2025 – Jan 2026

Cairo, Egypt (Remote) | *Flutter, Firebase*

- Built the complete Flutter frontend of the Driver & Medic mobile app, every screen from start to finish, with real-time GPS telemetry and live trip state for logistics coordination.
- Implemented trip-lifecycle workflows and post-trip EMS reporting modules for data integrity.

Junior Manager — AIESEC Oct 2025 – Mar 2026

Bengaluru, India | *Managed Global Volunteer operations, tracked departmental KPIs, led cross-functional teams.*

EDUCATION

Integrated B.Sc. in Computer Science — BITS Pilani 2024 – 2028

CGPA: 7.64 | *Coursework: Algorithms, Data Structures, Databases, Operating Systems, Networks, System Design, Statistics, Deep Learning*

CERTIFICATIONS

Machine Learning: Advanced Learning Algorithms — Stanford / DeepLearning.AI • Supervised ML — Stanford / DeepLearning.AI • Intermediate ML — Kaggle

LLM & GenAI: LangChain: Chat with Your Data (LangChain, DeepLearning.AI) • Building Systems with the ChatGPT API (OpenAI, DeepLearning.AI) • ChatGPT Prompt Engineering for Developers (OpenAI, DeepLearning.AI) • AI Fluency: Framework & Foundations (Anthropic)